SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement ID –SIH26003**
- **Problem Statement Title-** AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)
- **Theme-** MedTech / BioTech / HealthTech
- **PS Category- Software**
- **Team ID- AU/SIH/26-200**
- **Team Name- HexaForge**

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IDEA TITLE

OUR IDEA

An AI-powered platform designed to support elderly dementia patients with **cognitive training, memory assistance** and **daily activity support,** especially in remote areas of the North-Eastern Region.

Better Minds Brighter Days

OUR SOLUTION

- **Adaptive games** for memory, attention, recall, pattern & object recognition
- AI analyzes performance and **adjusts difficulty**
- **Voice-assisted multilingual** interaction with an elderly-friendly interface
- **Medicine, hydration, activity and appointment reminders**
- **Caregiver dashboard** for progress, activity and alerts
- **Offline-first** operation for low-connectivity areas

Support Engage Empower

OUR PROTOTYPE

A mobile/tablet application with:

- Patient-friendly **home screen**
- Interactive **cognitive games**
- AI-based **performance tracking** and **recommendations**
- **Voice interaction** and reminders
- **Offline local storage** with synchronization
- **Caregiver monitoring** dashboard

Technology with empathy for a brighter tomorrow

Elderly Empowered | Families Connected | Healthier Communities | Stronger North-East | *Together for a Better Tomorrow* 2

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TECHNICAL APPROACH

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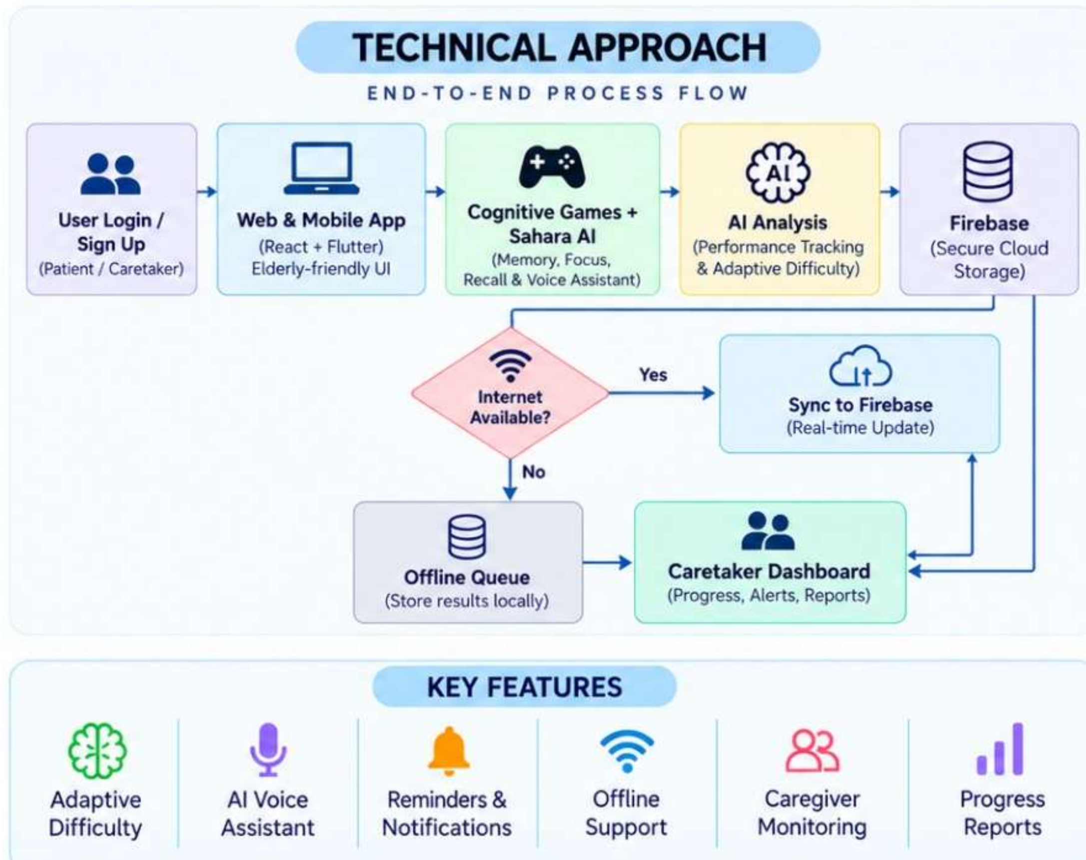

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FEASIBILITY AND VIABILITY

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ANALYSIS OF THE FEASIBILITY OF THE IDEA 01

Technical Feasibility
Built using proven web, AI/ML and cloud technologies (React, Firebase, Gemini AI).

Operational Feasibility
Works on smartphones, tablets and laptops with offline support.

Cost Feasibility
Uses cost-effective and scalable cloud services (Firebase, Gemini) with minimal infrastructure.

Scalability
Can be deployed across regions and scaled to more users and languages easily.

POTENTIAL CHALLENGES AND RISKS 02

Connectivity
Unstable or limited internet in remote areas.

Adoption
Elderly users may face digital, language or usability barriers.

Data Security
Sensitive patient information requires secure handling.

Model Limitations
AI responses may sometimes be inaccurate or need human oversight.

STRATEGIES FOR OVERCOMING THESE CHALLENGES 03

Offline-first Approach
Local storage with automatic sync when online.

Accessibility & Inclusivity
Simple, large UI with voice and multilingual support.

Security Measures
Firebase authentication and encrypted data handling.

Continuous Improvement
User feedback and performance data to refine AI and personalize experience.

VIABLE, SUSTAINABLE AND REAL-WORLD IMPACT

With the right technology, design and implementation, MindSathi is a **feasible, scalable** and **impactful** solution for improving the cognitive health and overall well-being of the elderly.

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IMPACT AND BENEFITS

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POTENTIAL IMPACT ON TARGET AUDIENCE ➡

- **Elderly Users** ⇨ Improved cognitive engagement, daily routine support and quality of life.
- **Caregivers & Families** ⇨ Easier progress monitoring and more timely support.
- **AI Personalization** ⇨ Activities adapt to individual cognitive performance and needs.

Happier Healthier Together

BENEFITS OF THE SOLUTION ➡

- **Social** ⇨ Better access, engagement and caregiver support for elderly users.
- **Economic** ⇨ Affordable digital support with reduced manual monitoring.
- **Environmental** ⇨ Reduces the need for frequent travel for routine cognitive support and monitoring.

Small Changes Big Impact

Elderly Empowered | Families Connected | Healthier Communities | Stronger North-East

Together for a Better Tomorrow

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RESEARCH AND REFERENCES

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- **Source:** Ministry of Development of North Eastern Region
- **National Institute on Aging (NIA)** – Cognitive Health & Dementia Resources
- **Alzheimer's & Related Disorders Society of India (ARDSI)** – Dementia Care Resources
- **Ministry of Development of North Eastern Region (MDoNER)** – Regional Development & Healthcare Resources
- **National Health Mission (NHM)** – Public Health & Healthcare Resources
- **Research Studies** – Cognitive Games, Memory Training & Digital Assistance for Elderly Users